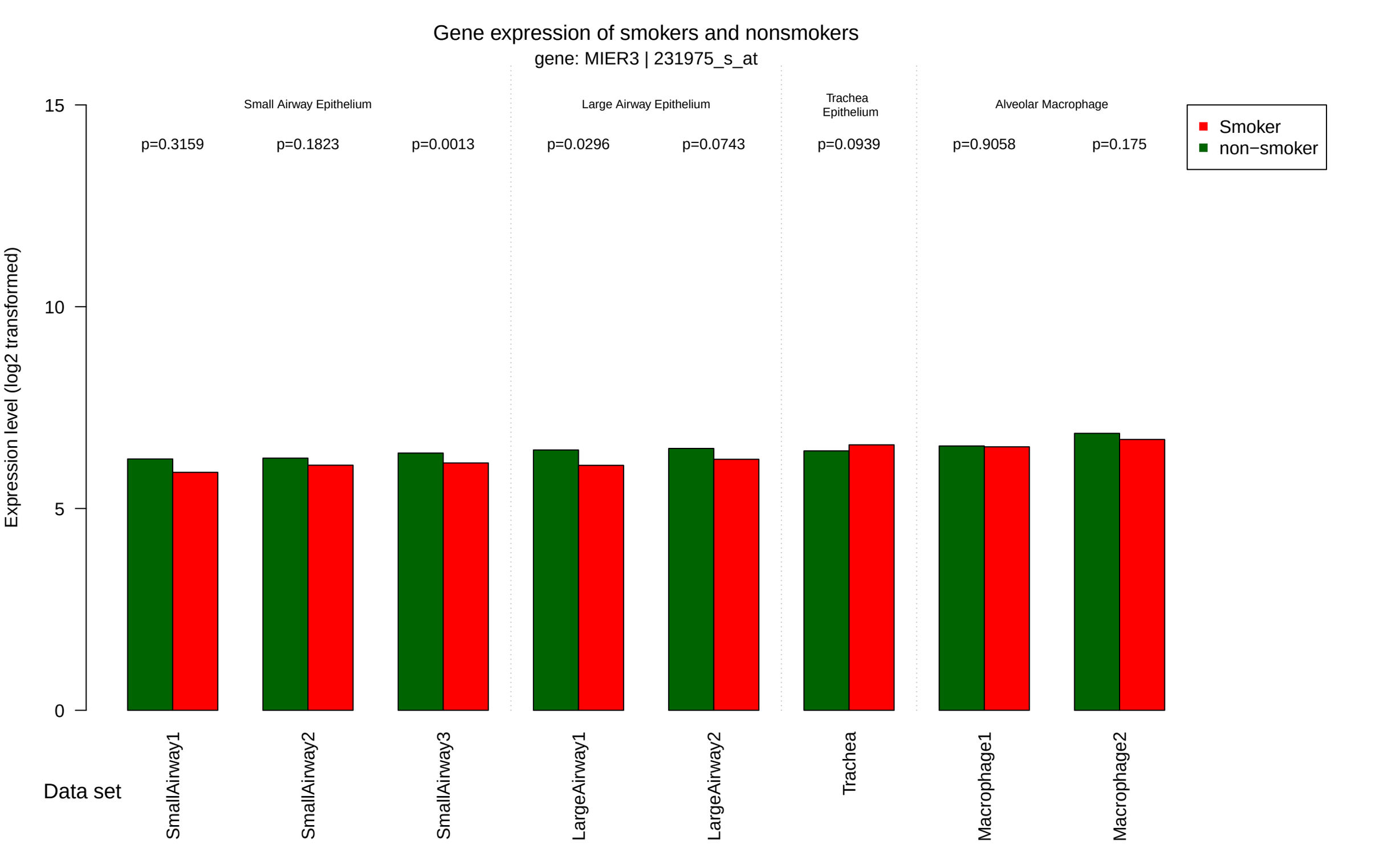
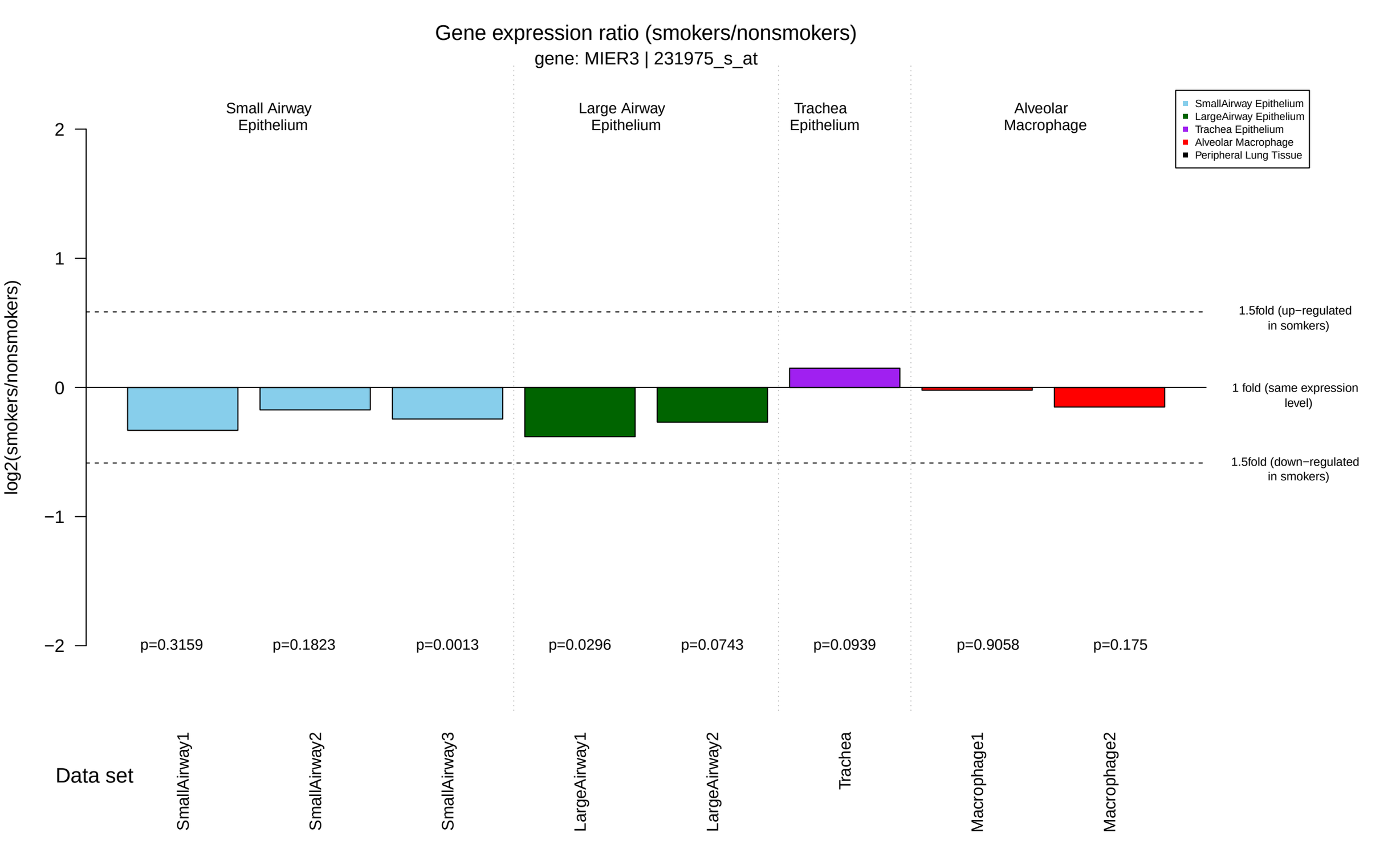
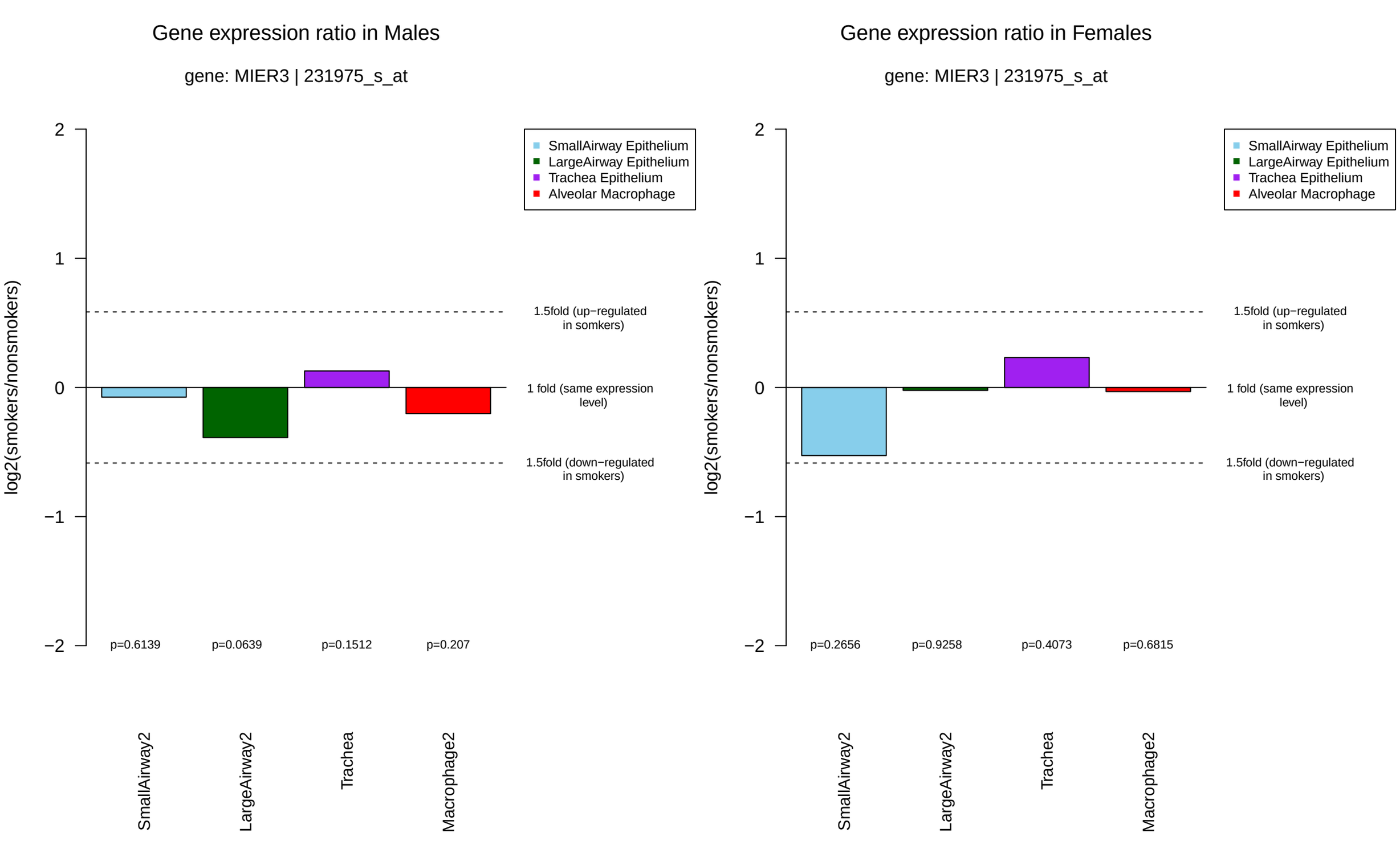
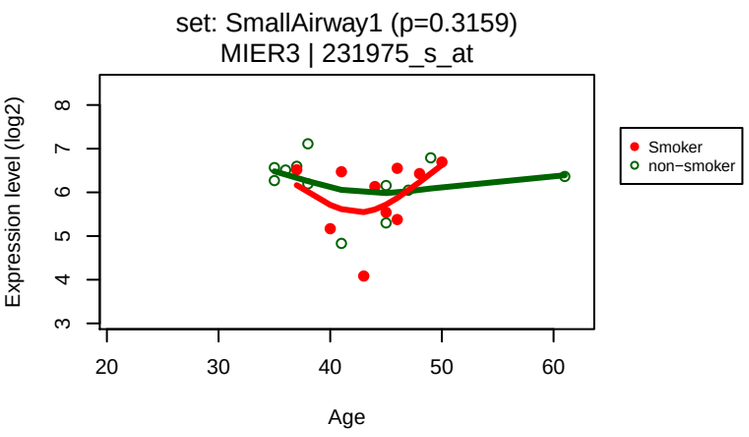
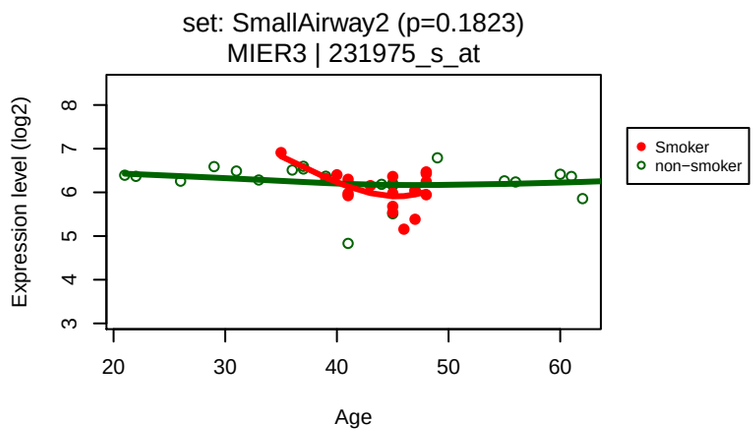
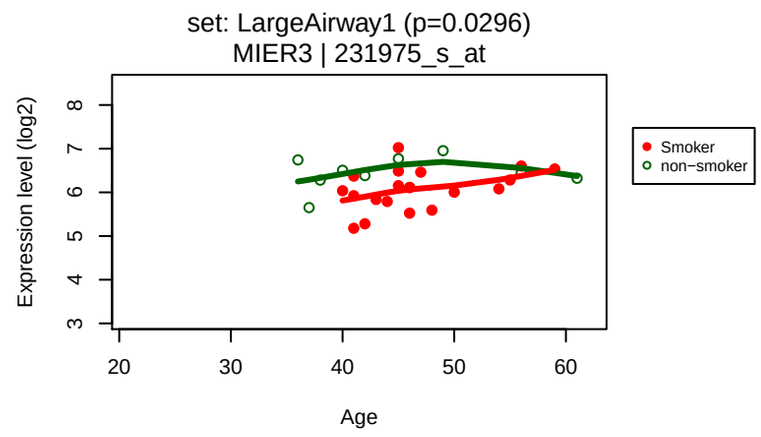

Gene expression of smokers and nonsmokers
gene: MIER3 | 231975_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3159
p=0.1823
p=0.0013
p=0.0296
p=0.0743
p=0.0939
p=0.9058
p=0.175
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

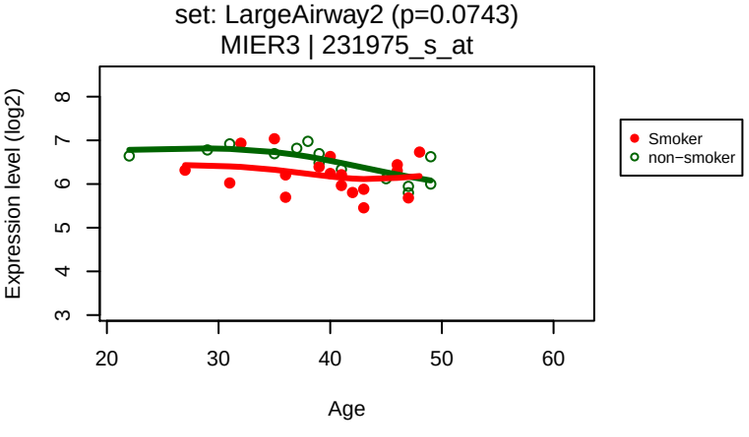

Gene expression ratio (smokers/nonsmokers)
gene: MIER3 | 231975_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3159
p=0.1823
p=0.0013
p=0.0296
p=0.0743
p=0.0939
p=0.9058
p=0.175
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

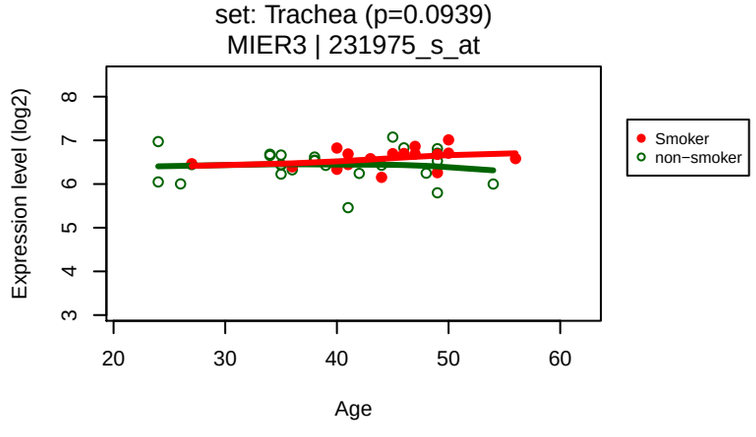

Gene expression ratio in Males
gene: MIER3 | 231975_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6139
p=0.0639
p=0.1512
p=0.207
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MIER3 | 231975_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2656
p=0.9258
p=0.4073
p=0.6815
SmallAirway2
LargeAirway2
Trachea
Macrophage2

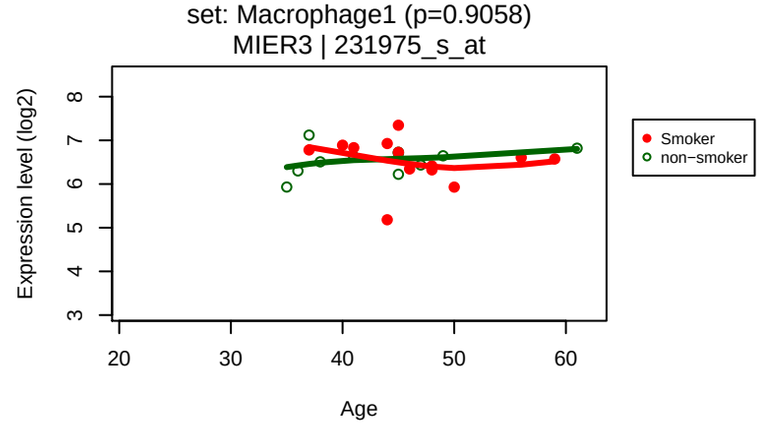

set: SmallAirway1 (p=0.3159)
MIER3 | 231975_s_at
Expression level (log2)
Age
Smoker
non−smoker

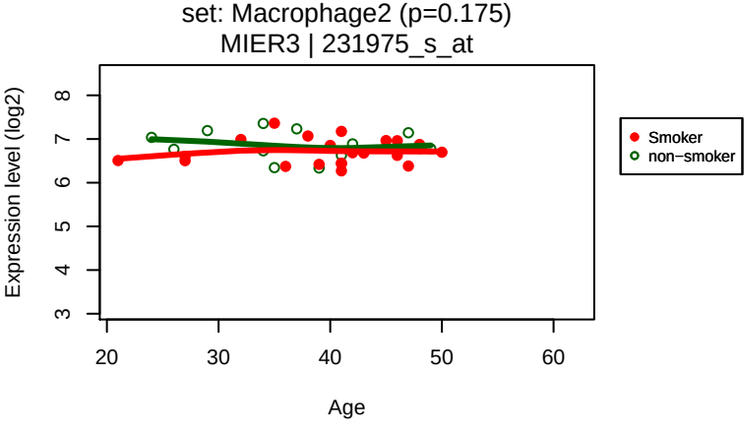

set: SmallAirway2 (p=0.1823)
MIER3 | 231975_s_at
Expression level (log2)
Age
Smoker
non−smoker

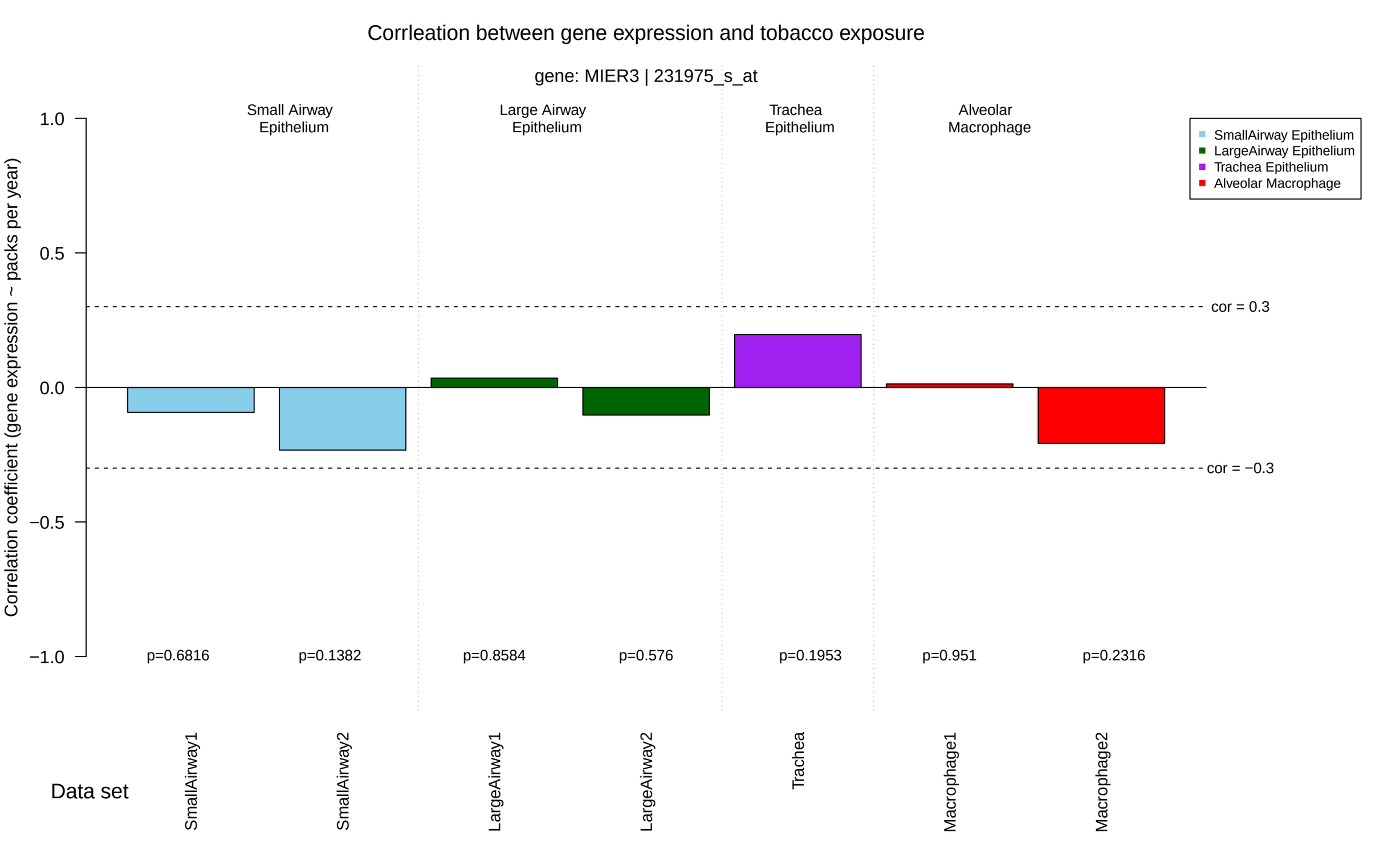

set: LargeAirway1 (p=0.0296)
MIER3 | 231975_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0743)
MIER3 | 231975_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0939)
MIER3 | 231975_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9058)
MIER3 | 231975_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.175)
MIER3 | 231975_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MIER3 | 231975_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6816
p=0.1382
p=0.8584
p=0.576
p=0.1953
p=0.951
p=0.2316
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2